

SOC L1 Log Investigation Practical Test (6 Scenarios)

Includes realistic SOC alerts, questions, options, and correct answers.

Scenario 1 — Authentication Log Analysis

SIEM Alert:

Alert Name: Multiple Failed Login Attempts

Source IP: 185.221.10.45

Target User: administrator

Event ID: 4625

Attempts: 320

Time Window: 4 minutes

1. What type of attack is likely occurring?

- A. Credential Stuffing
- B. Brute Force Attack
- C. Phishing
- D. SQL Injection

Correct Answer: B

2. Which log field indicates the attacker's source?

- A. Account Name
- B. Source Network Address
- C. Event ID
- D. Logon Type

Correct Answer: B

3. What does Event ID 4625 represent?

- A. Successful Login
- B. Failed Login Attempt
- C. Account Created
- D. Password Changed

Correct Answer: B

4. What should the SOC analyst check next?

- A. Successful login events (4624)
- B. Email logs

C. Firewall firmware

D. DNS records

Correct Answer: A

Scenario 2 — Suspicious Login Alert

SIEM Alert:

User: john.doe

Previous Login: India

New Login: Russia

Time Difference: 2 minutes

Authentication: O365

5. *What security indicator does this represent?*

A. Impossible Travel

B. Phishing

C. Malware

D. Data Exfiltration

Correct Answer: A

6. *What attack might have happened before this login?*

A. Phishing or credential theft

B. Hardware failure

C. DNS misconfiguration

D. Software update

Correct Answer: A

7. *What should the SOC analyst check immediately?*

A. Login history and IP reputation

B. CPU temperature

C. Printer logs

D. Monitor resolution

Correct Answer: A

8. *What immediate response action should be taken?*

A. Ignore alert

B. Force password reset and revoke sessions

C. Restart router

D. Delete user account

Correct Answer: B

Scenario 3 — Email Security Alert

Email Gateway Alert:

Sender: security-update@microsoft-support.com

Recipient: hr@company.com

Attachment: Salary_Adjustment_2025.xlsm

SPF: Fail

DKIM: Fail

DMARC: Fail

9. What type of attack is this?

- A. Phishing
- B. Ransomware
- C. DDoS
- D. Brute Force

Correct Answer: A

10. Which indicator suggests the email is malicious?

- A. SPF/DKIM/DMARC failures
- B. Correct grammar
- C. Known sender
- D. Internal email

Correct Answer: A

11. What risk does the .xlsm attachment indicate?

- A. Image file
- B. Macro-based malware
- C. Video file
- D. Audio file

Correct Answer: B

12. What should SOC do if the user opened the attachment?

- A. Ignore alert
- B. Isolate endpoint and scan system
- C. Delete email only
- D. Disable WiFi

Correct Answer: B

Scenario 4 — Endpoint Security Alert

EDR Alert:

Host: HR-Laptop-14

Process: mimikatz.exe

Target Process: lsass.exe

13. What attack technique is being used?

- A. Credential Dumping
- B. SQL Injection
- C. Cross-Site Scripting
- D. DDoS

Correct Answer: A

14. What sensitive data is the attacker trying to extract?

- A. Password hashes and Kerberos tickets
- B. Images
- C. Video files
- D. Printer drivers

Correct Answer: A

15. Which Windows process stores authentication credentials?

- A. explorer.exe
- B. lsass.exe
- C. notepad.exe
- D. chrome.exe

Correct Answer: B

16. What attack stages may follow this activity?

- A. Privilege escalation and lateral movement
- B. Backup operation
- C. Software update
- D. System reboot

Correct Answer: A

Scenario 5 — Lateral Movement Alert

SIEM Alert:

Source Host: HR-Laptop-14

Destination Host: FileServer-02

Protocol: RDP

User: Administrator

Time: 03:12 AM

17. What attack phase does this indicate?

- A. Lateral Movement
- B. Phishing
- C. Data Backup
- D. System Patch

Correct Answer: A

18. Why is this login suspicious?

- A. Admin account used from user workstation
- B. Normal business activity
- C. Scheduled task
- D. System update

Correct Answer: A

19. What previous attack likely enabled this?

- A. Credential dumping or phishing
- B. Hardware failure
- C. DNS update
- D. Antivirus scan

Correct Answer: A

20. What logs should SOC investigate next?

- A. Windows security logs
- B. Music files
- C. Printer logs
- D. Monitor settings

Correct Answer: A

Scenario 6 — Data Exfiltration Alert

Firewall Log:

Source IP: 192.168.10.15

Destination IP: 45.77.123.66

Protocol: HTTPS

Data Transfer: 4.8 GB

Time: 02:41 AM

Host: Finance-PC-03

21. What suspicious activity could this represent?

- A. Data Exfiltration
- B. Software update
- C. Backup process
- D. Antivirus scan

Correct Answer: A

22. *Why is the time of activity important?*

- A. Occurred outside normal business hours
- B. System reboot
- C. Hardware check
- D. Printer job

Correct Answer: A

23. *What should the SOC analyst investigate next?*

- A. File access logs and user activity
- B. Screen brightness
- C. Mouse driver
- D. Keyboard settings

Correct Answer: A

24. *What type of incident might this become?*

- A. Data breach
- B. Hardware issue
- C. Printer failure
- D. Software update

Correct Answer: A